

Weighted least-squares FIR design:

- The weighted-LS criterion allows more control over ripple performance:

$$\mathcal{E}_W = \frac{1}{2\pi} \int_{-\pi}^{\pi} W(\omega) |D(e^{j\omega}) - H(e^{j\omega})|^2 d\omega.$$

- For length- L GLP, $D(e^{j\omega}) = D(\omega)e^{-j(\omega\frac{L-1}{2} + \phi_0)}$, so

$$\begin{aligned} \mathcal{E}_W &= \frac{1}{2\pi} \int_{-\pi}^{\pi} W(\omega) |D(\omega)e^{-j(\omega\frac{L-1}{2} + \phi_0)} - H(e^{j\omega})|^2 d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} W(\omega) |D(\omega) - H(e^{j\omega})e^{j(\omega\frac{L-1}{2} + \phi_0)}|^2 d\omega. \end{aligned}$$

- For Type-I GLP filters (odd L , $\phi_0 = 0$, symmetric $h[n]$):

$$\begin{aligned} H(e^{j\omega})e^{j(\omega\frac{L-1}{2} + \phi_0)} &= \sum_{n=0}^{L-1} h[n]e^{-j\omega(n - \frac{L-1}{2})} \\ &= (h[0] + h[L-1])\cos(\omega\frac{L-1}{2}) + j(h[0] - h[L-1])\sin(\omega\frac{L-1}{2}) \\ &\quad + (h[1] + h[L-2])\cos(\omega\frac{L-3}{2}) + j(h[1] - h[L-2])\sin(\omega\frac{L-3}{2}) \\ &\quad + \dots + h[(L-1)/2] \\ &= h[(L-1)/2] + \sum_{n=0}^{\frac{L-1}{2}-1} 2h[n]\cos(\omega(\frac{L-1}{2} - n)) \quad \text{for real-valued } h[n] \\ &= \sum_{n=0}^{\frac{L-1}{2}} c_n \cos(\omega(\frac{L-1}{2} - n)) \quad \text{with } c_n = \begin{cases} 2h[n] & \text{for } n = 0 \dots \frac{L-1}{2} - 1 \\ h[n] & \text{for } n = \frac{L-1}{2} \end{cases} \end{aligned}$$

parametric model of
freq response

Weighted least-squares FIR design (cont.):

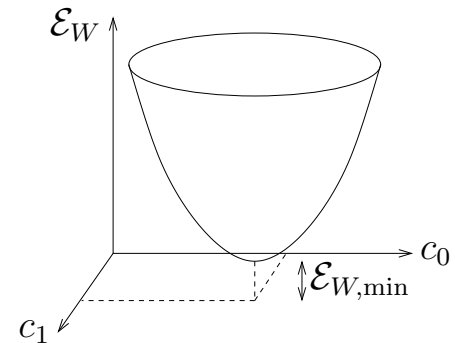
For real length- L Type-I GLP filters, the WLS criterion is:

$$\mathcal{E}_W = \frac{1}{2\pi} \int_{-\pi}^{\pi} W(\omega) \left[D(\omega) - \sum_{n=0}^{(L-1)/2} c_n \cos\left(\omega\left(\frac{L-1}{2} - n\right)\right) \right]^2 d\omega$$

function of $\begin{bmatrix} c_0 \\ \vdots \\ c_{\frac{L-1}{2}} \end{bmatrix}$

which is a quadratic function of the coefficients $\{c_n\}_{n=0}^{(L-1)/2}$.

At the optimal solution, the derivatives $\left\{ \frac{\partial \mathcal{E}_W}{\partial c_l} \right\}_{l=0}^{(L-1)/2}$ must equal zero.



These derivatives are:

$$\begin{aligned} \frac{\partial \mathcal{E}_W}{\partial c_l} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} W(\omega) \frac{\partial}{\partial c_l} \left[D(\omega) - \sum_{n=0}^{(L-1)/2} c_n \cos\left(\omega\left(\frac{L-1}{2} - n\right)\right) \right]^2 d\omega \\ &= \frac{-2}{2\pi} \int_{-\pi}^{\pi} W(\omega) \cos\left(\omega\left(\frac{L-1}{2} - l\right)\right) \left[D(\omega) - \sum_{n=0}^{(L-1)/2} c_n \cos\left(\omega\left(\frac{L-1}{2} - n\right)\right) \right] d\omega \\ &= -\frac{1}{\pi} \int_{-\pi}^{\pi} W(\omega) \cos\left(\omega\left(\frac{L-1}{2} - l\right)\right) D(\omega) d\omega \\ &\quad + \sum_{n=0}^{(L-1)/2} c_n \frac{1}{\pi} \int_{-\pi}^{\pi} W(\omega) \cos\left(\omega\left(\frac{L-1}{2} - l\right)\right) \cos\left(\omega\left(\frac{L-1}{2} - n\right)\right) d\omega. \end{aligned}$$

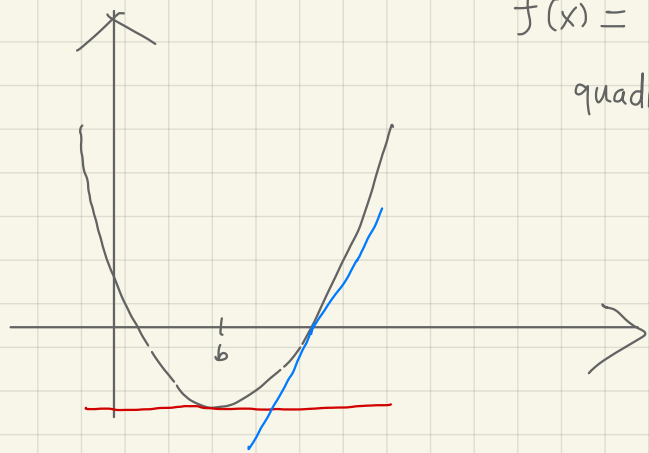
Zeroing gives $\frac{L-1}{2} + 1$ linear equations in $\frac{L-1}{2} + 1$ unknowns:

$$0 = -p_l + \sum_{n=0}^{(L-1)/2} c_n Q_{l,n} \text{ for } l = 0 \dots \frac{L-1}{2} \Leftrightarrow \mathbf{p} = \mathbf{Q}\mathbf{c}$$

which can be solved via $\mathbf{c} = \mathbf{Q}^{-1}\mathbf{p}$.

$$f(x) = a(x-b)^2 + c \quad a > 0$$

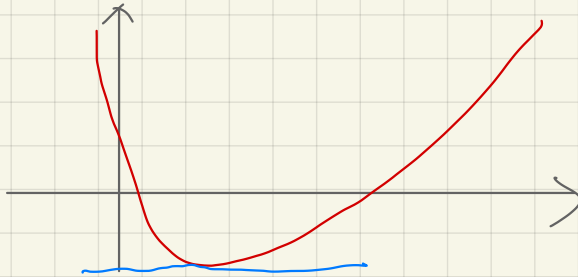
quadratic function



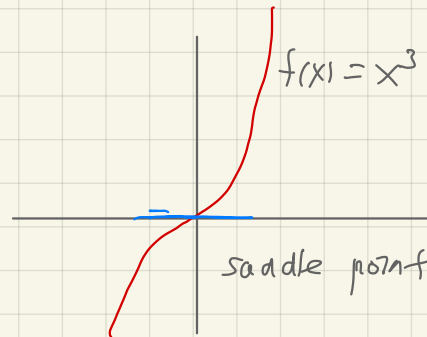
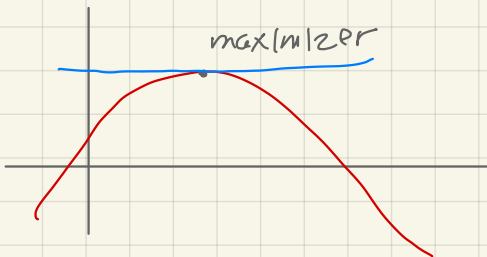
$$f'(x) = \lim_{\Delta \rightarrow 0} \frac{f(x+\Delta) - f(x)}{\Delta}$$

$$f'(x) = 2a(x-b)$$

$$f'(b) = 0 \quad f''(x) = 2a$$



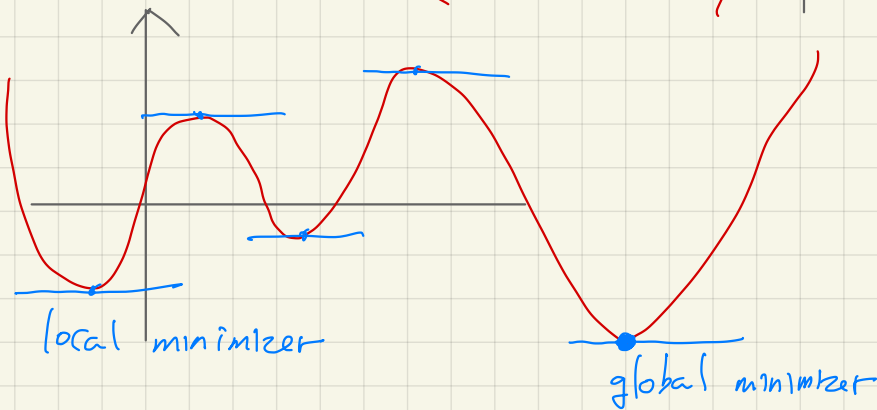
$$f'(x) = 0 ?$$



$$f(x) = x^3$$

$$f'(x) = 3x^2$$

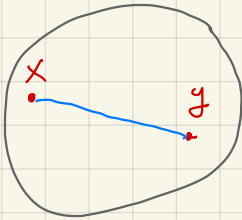
$$f'(0) = 0$$



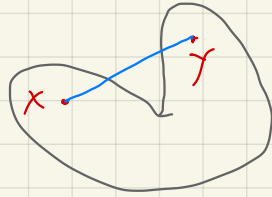
Convexity

S is a convex set if $\lambda x + (1-\lambda)y \in S$ for all $0 \leq \lambda \leq 1$ all $x, y \in S$
convex combination.

$S \in \mathbb{R}^2$



convex

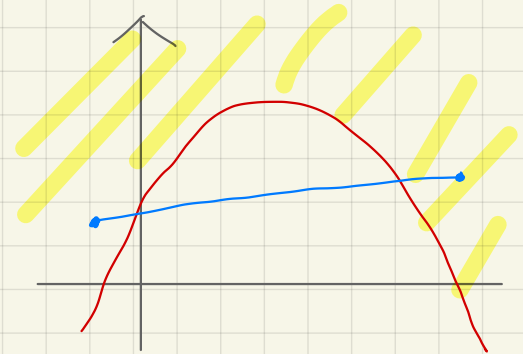
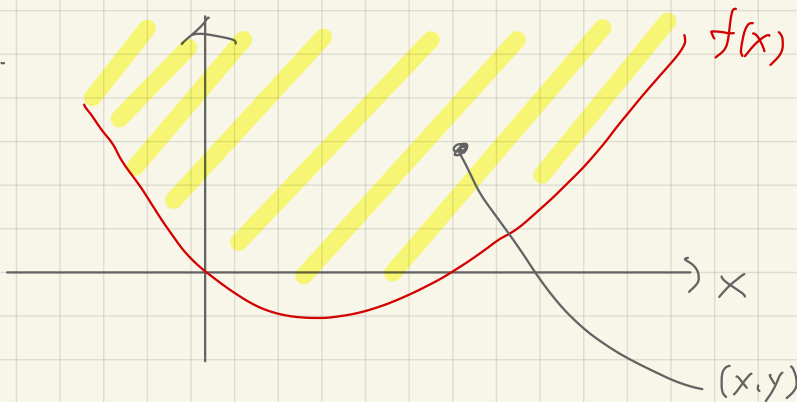


not convex

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$\text{epi}(f) = \{ (x, y) : y \geq f(x) \}$$

epigraph.



f is a convex function iff $\text{epi}(f)$ is a convex set.

If f is a convex function, and differentiable, then
 x_0 is a minimizer if $f'(x_0) = 0$

If f is twice differentiable,

f is convex if $f''(x) \geq 0$ for all x .

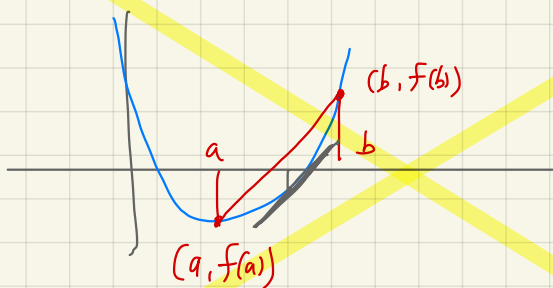
$$f: \mathbb{R}^d \rightarrow \mathbb{R}$$

d-variate function,

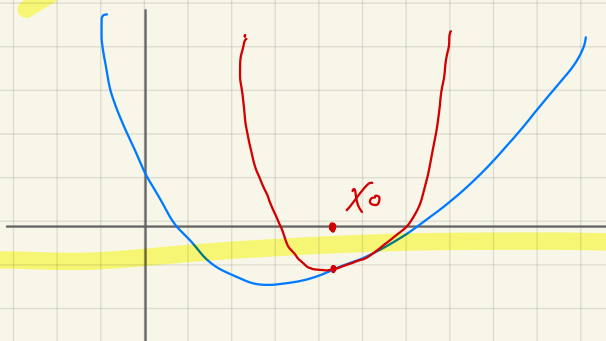
$d=1$

mean value theorem

$$\frac{f(b) - f(a)}{b - a} = f'(c) \quad \exists c \in [a, b]$$



$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(c)}{2}(x - x_0)^2$$



$$\nabla f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_d} \end{bmatrix}$$

gradient.

$$H_f(x) = \left[\frac{\partial^2 f(x)}{\partial x_i \partial x_j} \right]_{i,j}$$

Hessian matrix

$H_f(x)$ is positive semi definite (all eigenvalues are nonnegative)

then f is a convex function.

minimizer x_0 satisfies $\nabla f(x_0) = 0$

linear least squares.

$$\vec{y} = A\vec{x} + \vec{w}$$

observed $\vec{y}$ given $\vec{x}$ unknown signal A unknown noise $\vec{w}$

$$A \in \mathbb{R}^{m \times n}$$

$$\vec{y} \in \mathbb{R}^m \quad \vec{w} \in \mathbb{R}^m$$

$$\vec{x} \in \mathbb{R}^n$$

$$\min_{\vec{x}} \frac{1}{2} \|\vec{y} - A\vec{x}\|_2^2$$

$$f(\vec{x}) = \frac{1}{2} \|\vec{y} - A\vec{x}\|_2^2$$

$$\nabla f(\vec{x}) = A^T(A\vec{x} - \vec{y})$$

$$H = A^T A$$

$$\begin{aligned} \vec{z}^T H \vec{z} &= \vec{z}^T A^T A \vec{z} \\ &= \|A\vec{z}\|_2^2 \geq 0 \quad \forall \vec{z} \end{aligned}$$

$\exists$ $\vec{z}$ is an eigenvector of H with λ

$$H\vec{z} = \lambda\vec{z}$$

$$0 \leq \vec{z}^T H \vec{z} = \vec{z}^T \lambda \vec{z} = \lambda \underbrace{\|\vec{z}\|_2^2}_{>0}$$

$$\lambda \geq 0$$

H is positive semidefinite

(all eigenvalues are nonnegative)

$\Rightarrow f(\vec{x})$ is a convex function.

$$\nabla f(\vec{x}) = A^T(A\vec{x} - \vec{y}) = 0$$

$$\underbrace{A^T A} \vec{x} = A^T \vec{y}$$

$$\vec{x} = (A^T A)^{-1} A^T \vec{y}$$

normal equation.

geometric approach to least squares.

$$\min_{\vec{x}} \|\vec{y} - A\vec{x}\|_2^2$$

$$\begin{aligned} \min_{\vec{x}, \vec{z}} \|\vec{y} - \vec{z}\|_2^2 \\ \text{s.t. } \vec{z} = A\vec{x} \end{aligned}$$

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

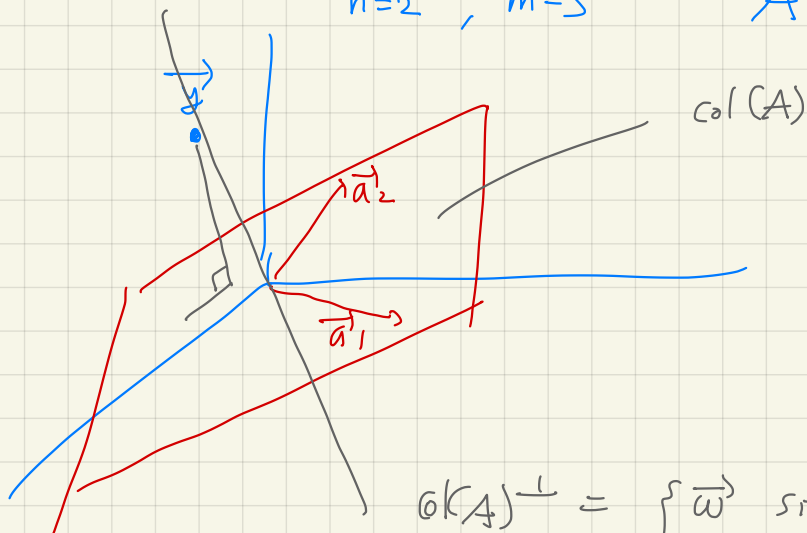
$$A = [\vec{a}_1 \dots \vec{a}_n]$$

$$A\vec{x} = \sum_{k=1}^n \vec{a}_k x_k$$

$\text{col}(A) = \{ A\vec{x} : \vec{x} \in \mathbb{R}^n \}$
column space of A

$n=2, m=3$

$$A \in \mathbb{R}^{3 \times 2}$$



$$\text{col}(A)^\perp = \{ \vec{w} \text{ s.t. } \vec{w}^T A\vec{x} = 0 \quad \forall \vec{x} \}$$

$$(\vec{y} - A\hat{\vec{x}}) \cdot A\vec{x} = 0 \quad \forall \vec{x}$$

$$A^T (A\hat{\vec{x}} - \vec{y}) = 0$$

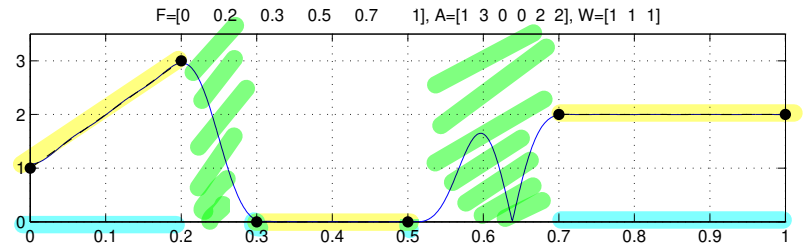
Example of weighted-LS FIR design:

Matlab's `firls(L-1,F,A,W)` performs WLS optimization:

F: frequencies (pairs)

A: amplitudes (pairs)

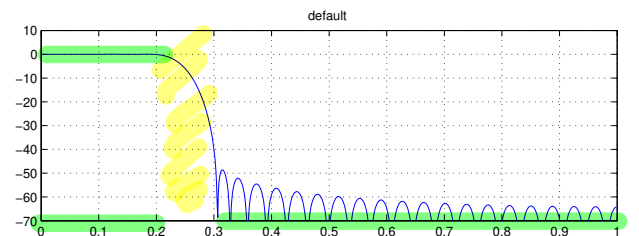
W: weights (singles)



The following code designs a WLS LPF and then tweaks it:

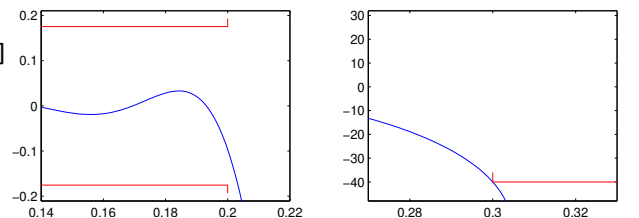
```
fp = 0.2; % matlab-normalized frequency!
fs = 0.3;
delp = 0.02;
dels = 0.01;
```

```
%L=53;
%F=[0,fp,fs,1]; A=[1,1,0,0]; W=[1/delp,1/dels];
%h=firls(L-1,F,A,W);
```

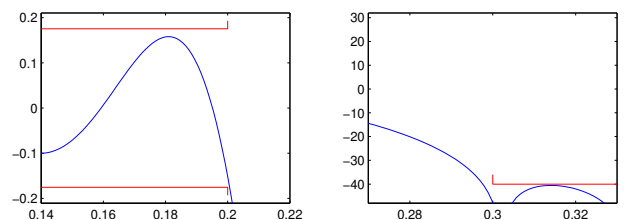
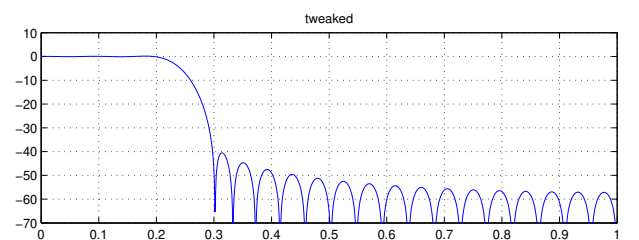


```
L=44;
F=[0,1.035*fp,0.975*fs,1]; A=[1,1,0,0]; W=[1/delp,2/dels]
h=firls(L-1,F,A,W);
```

```
Ap = -20*log10(1-delp);
As = -20*log10(dels);
```

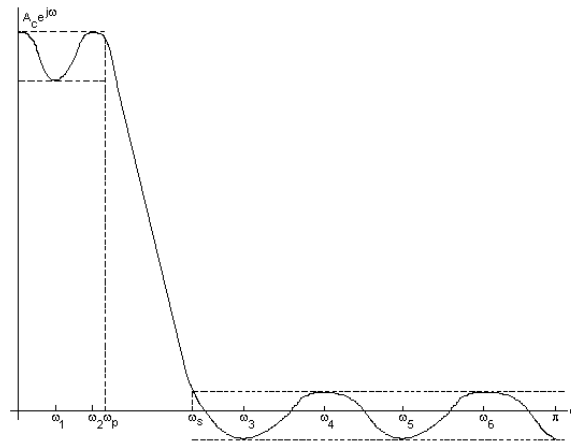


```
[H,omega]=freqz(h,1,1024);
f = omega/pi;
subplot(2,1,1);
plot(f,20*log10(abs(H)));
grid on;
axis([0,1,-70,10])
subplot(2,2,3);
plot(f,20*log10(abs(H)),...
     [0,fp,fp],-Ap*[1,1,1.1], 'r',...
     [0,fp,fp],Ap*[1,1,1.1], 'r');
axis([0.7*fp,1.1*fp,-1.2*Ap,1.2*Ap]);
subplot(2,2,4);
plot(f,20*log10(abs(H)),...
     [fs,fs,1],-As*[0.9,1,1], 'r');
axis([0.9*fs,1.1*fs,-1.2*As,0.8*As]);
```



Equiripple (Parks-McClellan) FIR design:

- If we want to minimize the peak deviation from the desired $D(\omega)$, then *equiripple* filters are optimal.
- Intuitively speaking, when we try to suppress one ripple, the others get larger. Imagine repeatedly re-designing a WLS filter with increased weight around the worst ripple. We would eventually get a filter with equal-sized ripples:



- The Parks-McClellan algorithm accomplishes this task, but in a much faster way, using the Chebyshev's alternation theorem and the Remez exchange algorithm.
- `firpm(L-1,F,A,W)` performs Matlab equiripple design, with the same interface as `firls(...)`.
- The command `firpmord` aids in the selection of the `firpm` input parameters.

GLP Type 1 (odd L , $\phi_0=0$, symmetric)

$$A(\omega) = H(e^{j\omega}) e^{j(\frac{L-1}{2}\omega + \phi_0)} \quad \begin{aligned} H(e^{j\omega}) &= h[0] + h[1]e^{-j\omega} + h[2]e^{-j2\omega} \\ &= e^{-j\omega} (h[0]e^{j\omega} + h[1] + h[2]e^{-j\omega}) \end{aligned}$$
$$= h\left[\frac{L-1}{2}\right] + 2 \sum_{n=1}^{\frac{L-1}{2}} h\left[\frac{L-1}{2} - n\right] \cos(n\omega)$$

trigonometric identities

$$\cos(2\omega) = -1 + 2\cos^2\omega$$

$$\cos(3\omega) = -3\cos\omega + 4\cos^3\omega$$

$$\cos(4\omega) = 1 - 8\cos^2\omega + 8\cos^4\omega$$

$$\cos(5\omega) = 5\cos\omega - 20\cos^3\omega + 16\cos^5\omega$$

In general $\cos(n\omega) = T_n(\cos\omega)$

$T_n(x)$ is a polynomial of order n

$$A(\omega) = \sum_{n=0}^{\frac{L-1}{2}} \alpha_n \cos^n(\omega)$$

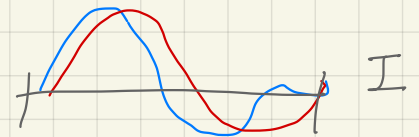
Chebyshev's alternating theorem.

Let $P(x) = \sum_{k=0}^M \alpha_k x^k$ M -th-order polynomial.

$D(x)$: desired target function

$W(x)$: positive continuous weight function

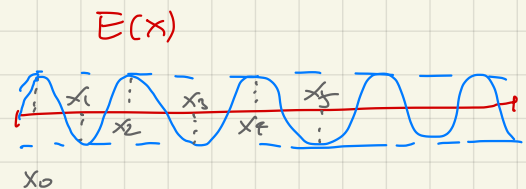
$$E(x) = W(x) [D(x) - P(x)]$$



obj: $\min_{\{\alpha_k\}} \max_{x \in I} |E(x)|$



$P(x)$ is optimal



iff

there exist $x_0 < x_1 < \dots < x_{M+1}$

such that

$$E(x_{j+1}) = -E(x_j)$$

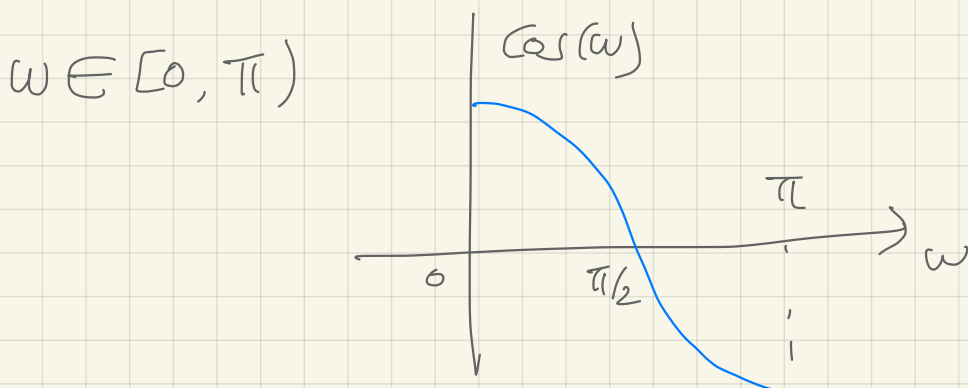
$\forall j$

and

$$|E(x_j)| = \max_{x \in I} |E(x)|$$

back to filter design.

$$\min_{\{\alpha_k\}} \max_{\omega \in I} \left| W(\omega) \left(D(\omega) - \sum_{k=0}^{\frac{L-1}{2}} \alpha_k \cos^k \omega \right) \right|$$



$x = \cos \omega$ change of variable

$$\min_{\{\alpha_k\}} \max_{x \in \cos(I)} \left| W(\cos^{-1} x) \left(D(\cos^{-1}(x)) - \sum_{k=0}^{\frac{L-1}{2}} \alpha_k x^k \right) \right|$$

optimality condition $\equiv \frac{L-1}{2} + 2$ alternations

$$x_0 < x_1 < \dots < x_{\frac{L+1}{2}}$$

$$W(\cos^{-1} x_j) \left[D(\cos^{-1}(x_j)) - \sum_{k=0}^{\frac{L-1}{2}} \alpha_k (x_j)^k \right] = (-1)^j f$$

$$\sum_{k=0}^{\frac{L-1}{2}} \alpha_k \cos^k \omega_j = h\left[\frac{L-1}{2}\right] + \sum_{k=1}^{\frac{L-1}{2}} h\left[\frac{L-1}{2} - k\right] \cos(k\omega_j)$$

alternations become

$$h\left[\frac{L-1}{2}\right] + \sum_{k=1}^{\frac{L-1}{2}} 2h\left[\frac{L-1}{2} - k\right] \cos(k\omega_j) + \frac{(-1)^j f}{W(\omega_j)} = D(\omega_j)$$

$$\begin{bmatrix} 1 & 2\cos(\omega_0) & 2\cos(2\omega_0) & \dots & 2\cos\left(\frac{L-1}{2}\omega_0\right) \\ 1 & 2\cos(\omega_1) & 2\cos(2\omega_1) & \dots & 2\cos\left(\frac{L-1}{2}\omega_1\right) \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 2\cos(\omega_{\frac{L-1}{2}}) & 2\cos(2\omega_{\frac{L-1}{2}}) & \dots & 2\cos\left(\frac{L-1}{2}\omega_{\frac{L-1}{2}}\right) \end{bmatrix} \begin{bmatrix} \frac{1}{W(\omega_0)} \\ -\frac{f}{W(\omega_1)} \\ \vdots \\ h[0] \\ f \end{bmatrix} = \begin{bmatrix} D(\omega_0) \\ D(\omega_1) \\ \vdots \\ D(\omega_{\frac{L-1}{2}}) \end{bmatrix}$$

need to find

$$0 < \omega_0 < \omega_1 < \dots < \omega_{\frac{L-1}{2}} < \pi$$

$$h[0], h[1], \dots, h\left[\frac{L-1}{2}\right]$$

f

Alternating method

Initial guess on

$$0 < \omega_0 < \omega_1 < \dots < \omega_{\frac{L+1}{2}} < \pi$$

compute $\delta, h[0], h[1], \dots, h[\frac{L-1}{2}]$
by least squares to

$$\begin{bmatrix} 1 & 2\cos(\omega_0) & 2\cos(2\omega_0) & \dots & 2\cos(\frac{L-1}{2}\omega_0) \\ 1 & 2\cos(\omega_1) & 2\cos(2\omega_1) & \dots & 2\cos(\frac{L-1}{2}\omega_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2\cos(\omega_{\frac{L+1}{2}}) & 2\cos(2\omega_{\frac{L+1}{2}}) & \dots & 2\cos(\frac{L-1}{2}\omega_{\frac{L+1}{2}}) \end{bmatrix} \begin{bmatrix} \frac{1}{W(\omega_0)} \\ \frac{1}{W(\omega_1)} \\ \vdots \\ \frac{1}{W(\omega_{\frac{L+1}{2}})} \end{bmatrix} \begin{bmatrix} h[\frac{L-1}{2}] \\ h[\frac{L-3}{2}] \\ \vdots \\ h[0] \\ \delta \end{bmatrix} = \begin{bmatrix} D(\omega_0) \\ D(\omega_1) \\ \vdots \\ D(\omega_{\frac{L+1}{2}}) \end{bmatrix}$$

$$A(\omega) = h[\frac{L-1}{2}] + 2 \sum_{n=1}^{\frac{L-1}{2}} h[\frac{L-1}{2} - n] \cos(n\omega)$$

$$E(\omega) = W(\omega) [D(\omega) - A(\omega)]$$

Find new $\frac{L+1}{2}$ peak error locations

Remez exchange algorithm.

Initial guess on

$$0 < \omega_0 < \omega_1 < \dots < \omega_{\frac{L+1}{2}} < \pi$$

Compute δ by

$$\delta = \frac{\sum_{j=0}^{\frac{L+1}{2}} \gamma_j D(\omega_j)}{\sum_{j=0}^{\frac{L+1}{2}} \frac{(-1)^j \gamma_j}{W(\omega_j)}}$$

$$\gamma_j = \prod_{l \neq j} \frac{1}{(\cos(\omega_j) - \cos(\omega_l))}$$

$$A(\omega_j) = \frac{(-1)^j}{W(\omega_j)} \delta + D(\omega_j) \quad \forall j$$

Lagrange interpolation to $\hat{A}(\omega)$

Find new $\frac{L+1}{2}$ peak error locations

Lagrange Interpolation

$$p(x) = \sum_{k=0}^M \alpha_k x^k$$

$$p(x) = \sum_{k=0}^M p(x_k) \frac{\prod_{\ell \neq k} (x - x_\ell)}{\prod_{\ell \neq k} (x_k - x_\ell)}$$

It suffices to know $M+1$ distinct samples.

Example of equiripple FIR design:

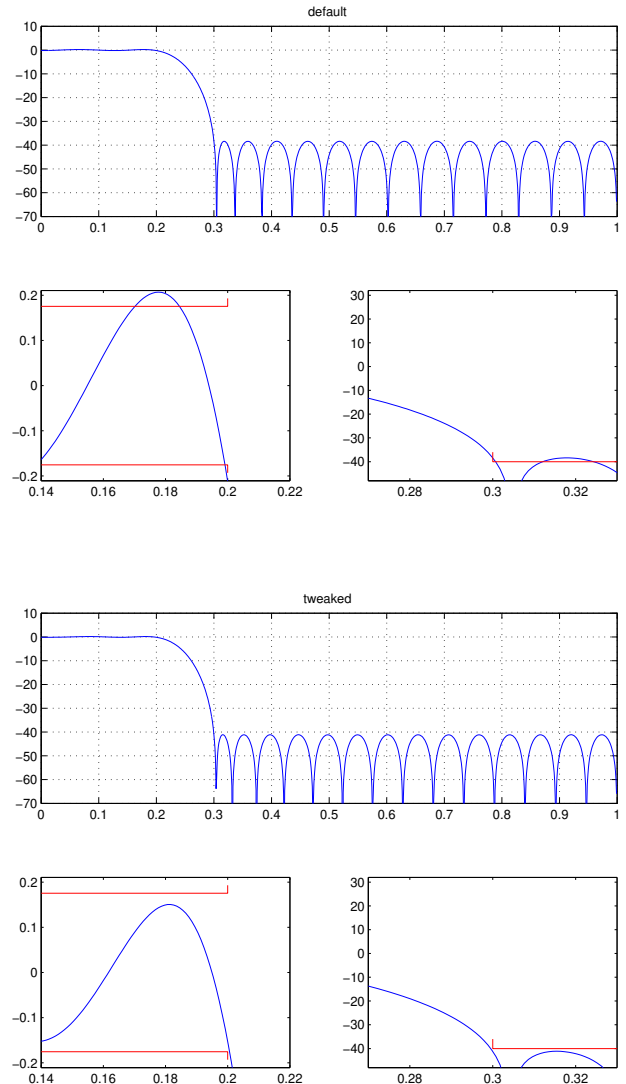
The following code designs an equiripple LPF using `firpm` with inputs generated using `firpmord`.

```
fp = 0.2; % matlab-normalized frequency!
fs = 0.3;
delp = 0.02;
dels = 0.01;

[N,F,A,W]=firpmord([fp,fs],[1,0],[delp,dels])
%h=firpm(N,F,A,W);
h=firpm(N+2,F,A,W);

Ap = -20*log10(1-delp);
As = -20*log10(dels);

[H,omega]=freqz(h,1,1024);
f = omega/pi;
subplot(2,1,1);
    plot(f,20*log10(abs(H)));
    grid on;
    axis([0,1,-70,10])
subplot(2,2,3);
    plot(f,20*log10(abs(H)),...
        [0,fp,fp],-Ap*[1,1,1.1],'r',...
        [0,fp,fp],Ap*[1,1,1.1],'r');
    axis([0.7*fp,1.1*fp,-1.2*Ap,1.2*Ap]);
subplot(2,2,4);
    plot(f,20*log10(abs(H)),...
        [fs,fs,1],-As*[0.9,1,1],'r');
    axis([0.9*fs,1.1*fs,-1.2*As,0.8*As]);
```



The order $N = 35$ guessed by `firpmord` was a bit too small, and so it was increased by 2 to meet the specs.

Summary of FIR filter design:

For GLP FIR filter design, we examined:

design method	matlab command(s)	example length L
window-based	fir1	60 (Hann)
	kaiserord	46 (Kaiser)
freq-sampled	fir2	68 (Hann)
weighted least-squares	firls	44
equiripple	firpm firpmord	38

where the “example length” was the minimum required to design a particular LPF under peak ripple constraints.